# Sre Cibi Varsshan K

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## EDUCATION

**Vellore Institute of Technology, Chennai** | Expected 2027
B.Tech Computer Science and Engineering (Artificial Intelligence and Machine Learning)
Relevant coursework: Machine Learning, Computer Vision, Data Structures and Algorithms.

## SUMMARY

B.Tech AI and ML student working on reinforcement learning and applied ML: a from-scratch TD3 agent with recorded experiments, a pixel-based DQN, image-classification comparisons, explainable retrieval, and on-device LLM inference on Android.

## PROJECTS

**TrackMania TD3 - reinforcement learning research** | Python, PyTorch, TMRL, pytest | Sep 2026
github.com/thenamelessone888/RL

- Implemented TD3 from scratch on the TMRL real-time framework for TrackMania 2020: twin critics with clipped double-Q targets, delayed policy updates, target policy smoothing and Polyak-averaged targets, on LIDAR and speed observations.
- Recorded and inspected human driving demonstrations (about 5,000 transitions) and behaviour-cloned the actor as a warm start before off-policy training.
- Diagnosed a policy collapse right after critic warm-up and fixed it with TD3+BC-style regularisation toward the cloned actor, with a decaying anchor weight; an extended run then held without collapse but plateaued.
- Root-caused the plateau to a recurring wall collision from deterministic evaluation traces and implemented a stuck-recovery reward penalty; built curriculum-learning stage and promotion tooling and 13 offline pytest tests.

**Space Shooter DQN (coursework)** | Python, PyTorch, Pygame | Apr 2026
github.com/xerfzy/Space_shooter_using_DQN_vc

- Trained a Dueling Double DQN that plays a custom Pygame shooter from raw pixels: four stacked 84x84 grayscale frames into a Nature-DQN CNN with separate value and advantage streams.
- Used a 50,000-transition replay buffer, Huber loss, gradient-norm clipping, epsilon-greedy decay, a periodically synced target network and wave-based reward shaping.

**XrivenixAI - on-device LLM assistant** | Kotlin, C++, llama.cpp, Room | Sep 2026
github.com/xerfzy/XrivenixAI

- Ran quantised GGUF models (1B to 7B, Q4_K_M) on Android CPUs through llama.cpp, with grammar-constrained extraction and KV-cache reuse; measured about 5-6 tokens/s generation for a 3B model on a Snapdragon 782G phone.
- Built hybrid retrieval for a long-term memory: bge-small embeddings, SQLite FTS5 and entity links merged into one score with per-route thresholds, evaluated with on-device retrieval tests and benchmarks.

**Breast Cancer Histopathology Classification (coursework)** | Python, TensorFlow, Keras | Nov 2025
github.com/xerfzy/Breast_cancer_calssification-vc-

- Compared CNN, ResNet, ViT and Swin Transformer classifiers on histopathology images split by magnification (40x to 400x); single-run test accuracy ranged from 0.68 (ResNet) to 0.92 (CNN, Swin).

**Legal Case Similarity with Explanations** | Python, Sentence-Transformers, FAISS, SHAP, LIME | Apr 2026
github.com/xerfzy/xai-v2

- Built a retrieval pipeline with bi-encoder embeddings, FAISS search, cross-encoder reranking and a NetworkX similarity graph, explained with attention weights, SHAP-style token scores and LIME; demonstrated on a small demo corpus with 37 pytest tests.

## RESEARCH

**Indian Patent Application IN202641060337 A1** (published application, not granted) | May 2026
"A System for Swarm of Drone for Crowed Behaviour Analysis". Application No. 202641060337, filed 12 May 2026, published 22 May 2026. Inventors: Sre Cibi Varsshan K, D Kavitha. Applicant: Vellore Institute of Technology, Chennai.

## TECHNICAL SKILLS

**Machine learning:** PyTorch, TensorFlow/Keras, reinforcement learning (TD3, DQN, behaviour cloning), CNNs, Vision Transformers, Sentence-Transformers, FAISS, SHAP, LIME, scikit-learn
**Data and tooling:** Python, NumPy, Pandas, Jupyter, pytest, Git, Linux
**Other:** C++, Kotlin, llama.cpp, OpenCV, Streamlit